Project Initialization and Planning Phase

Date	1 November 2026
Team ID	To be provided
Project Title	Global Olympic Sports Performance Analysis Using Tableau
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

A data-visualization project that explores Olympic athlete and medal records to identify country performance, sport-wise medal patterns, participation trends, host-nation performance, and (when supplementary data is available) sport investment and regional efficiency.	
Objective	To transform historical Olympic data into interactive Tableau visualizations, dashboards, and a story that help sports committees, funding analysts, and media use understand performance patterns and make evidence-informed decisions.
Scope	The project uses the uploaded athlete_events.csv dataset, which contains athlete-level Olympic records from 1896–2016. The analysis will focus primarily on 1992–2016, Summer and Winter Games, and nine visualizations across three dashboards. Addition investment, region, and host-country reference data are required for the full scenarios. Flask embedding is planned as a presentation layer.
Olympic performance data is large and spread across many years, countries, sports, and athlete records. Without suitable preparation and visualization, it is difficult for stakeholders to compare medal performance, understand participation patterns, identify strong sports, and communicate trends clearly.	
Description	The project addresses the difficulty of extracting meaningful insights from historical Olympic data by preparing the records, defining consistent medal metrics, and presenting interactive charts with filters.
Impact	The solution can make comparisons faster, reveal patterns that are difficult to see in raw tables, support evidence-based discussion of sports development, and communic Olympic trends to non-technical audiences. It does not prove that investment causes medal success.
Build a Tableau-based analytical solution with nine visualizations, three dashboards, and a four-scene story, supported by data cleaning, calculated fields, filters, and optional Flask web embedding.	
Approach	1) Import CSV data into Tableau. 2) Check fields, data types, missing values, duplicates and year coverage. 3) Create medal and athlete metrics. 4) Build nine charts for the three stakeholder scenarios. 5) Assemble dashboards and story. 6) Validate calculations and filters. 7) Publish/embed through Flask if hosting access is available.
Key Features	Interactive year, season, country, sport, and medal filters; country and sport comparisons; medal trend analysis; host-nation analysis using a host lookup; regional and investment analysis using supplementary datasets; three dashboards; four story points; and web presentation through Flask.

Resource Requirements

Resource Type	Description	Specification/Allocation

A standard laptop/desktop; modern multi-core CPU; GPU not required for Tableau-level aggregation.		
Hardware	Memory	Recommended 8 GB RAM minimum; 16 GB preferred for smoother work with larger extracts.
Hardware	Storage	At least 2 GB free space for CSV files, Tableau workbook, exports, and documentation; more if supplementary datasets are added.
Software	Frameworks	Tableau Public/Desktop for analysis; Python Flask for optional web integration.
Optional Python: pandas for preprocessing; Flask for web serving. Tableau calculations are used for visualization metrics.		
Software	Development Environment	Tableau Desktop/Public; Visual Studio Code or another code editor for Flask; web browser; Git optional for version control.
Data	Data Source, Size, Format	Uploaded athlete_events.csv; CSV format; 271,116 rows and 15 columns, covering 1896–2016. The file does not include investment or region fields; supplementary datasets are needed for those analyses.
Development Environment	IDE, version control	e.g., Jupyter Notebook, Git
Data		
Data	Source, size, format	e.g., Kaggle dataset, 10,000 images